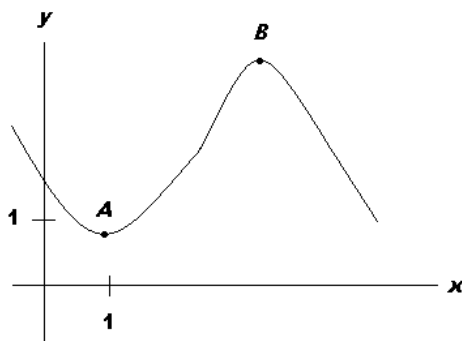


Homework Section 13.3

1. Find the exact value of the length of the curve given by $\mathbf{r}(t) = 2t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$, $-2 \leq t \leq 8$.
2. Set up the integral that gives the length of the following curve and then approximate that length, accurate to the nearest thousandth, using a calculator:
 $\mathbf{r}(t) = 2t\mathbf{i} + e^t\mathbf{j} - 3t^2\mathbf{k}$, $-5 \leq t \leq 6$.
3. Find the arc length function (starting from the point corresponding to $t = 0$) and use it to reparameterize $\mathbf{r}(t)$ in terms of arc length.
 - a) $\mathbf{r}(t) = 5t\mathbf{i} + \cos t\mathbf{j} + \sin t\mathbf{k}$
 - b) $\mathbf{r}(t) = \langle 3t, 1 + 2t, 4 - t \rangle$
4. Find the unit tangent and unit normal vectors:
 - a) $\mathbf{r}(t) = 2t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$
 - b) $\mathbf{r}(t) = e^{-t}\mathbf{i} + e^t\mathbf{j} - \sqrt{2}t\mathbf{k}$
5. Find the curvature:
 - a) $\mathbf{r}(t) = 2t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$
 - b) $\mathbf{r}(t) = e^{-t}\mathbf{i} + e^t\mathbf{j} - \sqrt{2}t\mathbf{k}$
6. Find the curvature: $\mathbf{r}(t) = 2t\mathbf{i} + (1 - t^2)\mathbf{j} - 3t\mathbf{k}$.
7. Find the curvature of $\mathbf{r}(t) = 2t\mathbf{i} + (1 - t^2)\mathbf{j} - 3t\mathbf{k}$ at the point $(2, 0, -3)$.
8. Find the curvature of $y = 8x^4 - x$.
9.
 - a) Where does the below curve have greater curvature, point A or point B? Justify your answer.
 - b) Approximate the curvature at point B by drawing the osculating circle at B.



10. Find the unit tangent, normal, and binormal vectors of $\mathbf{r}(t) = \frac{1}{3}t^3\mathbf{i} - \frac{\sqrt{2}}{2}t^2\mathbf{j} - t\mathbf{k}$ at the point given by $t = 1$.
11. Find the equations of the normal plane and osculating plane of $\mathbf{r}(t) = \frac{1}{3}t^3\mathbf{i} - \frac{\sqrt{2}}{2}t^2\mathbf{j} - t\mathbf{k}$ at the point given by $t = 1$.